Arduino Uno: keypad and LCD interface

ME3202 Lab 6 study | Gajendra, 2301ME18

Abstract - short summary

This report documents the course keypad/LCD interface and a fresh Arduino sketch in the portfolio. A key press is intended to appear on a 16x2 I2C display and Serial Monitor. The diagrams show the wiring and logic to check; this repository copy has not been compiled or retested on hardware.

Purpose and background

A 4x4 matrix keypad reduces the number of required signal pins by sharing four rows and four columns. The I2C LCD uses SDA and SCL for display communication.

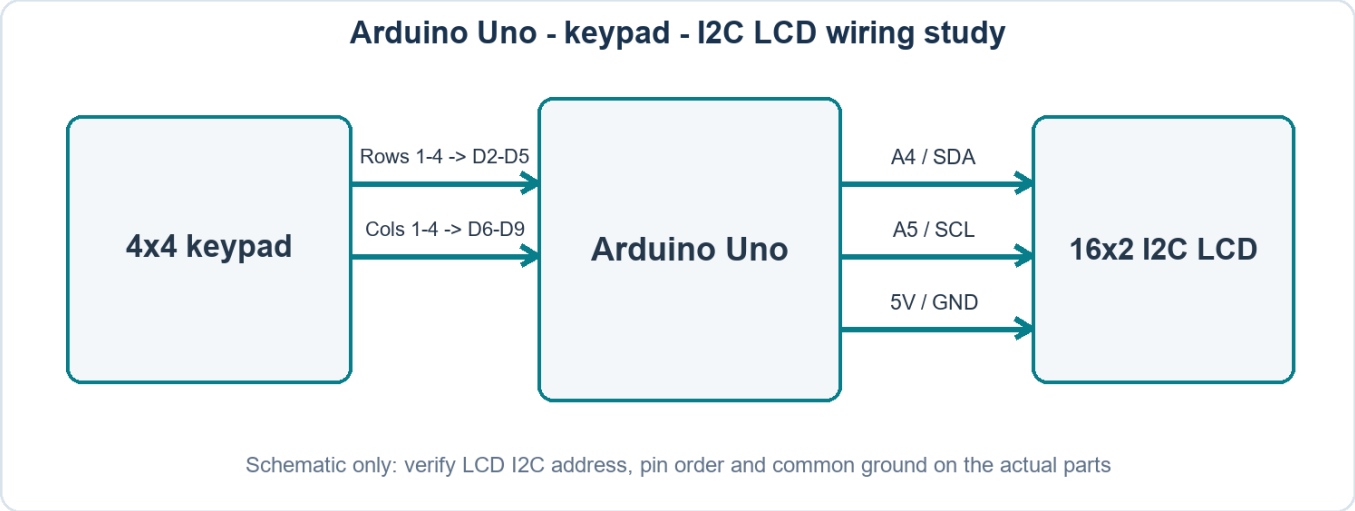


Figure 1. Pin mapping used by the fresh sketch. Confirm the particular keypad and LCD module pin order before wiring.

Methods - how the study was carried out

- Map keypad rows to D2-D5 and columns to D6-D9; connect LCD SDA to A4 and SCL to A5.
- Use the Keypad library to scan for a pressed key and LiquidCrystal_I2C to update the 16x2 display.
- Send the same key to Serial at 9600 baud for debugging; check all 16 keys on a physical setup.

Arduino Uno: keypad and LCD interface | Work and results

What was done

The items below describe the work represented by this repository and the course context.

• Gajendra reports completing the institute keypad/LCD lab.

• The repository includes a newly written .ino sketch and a documented pin table for practice.

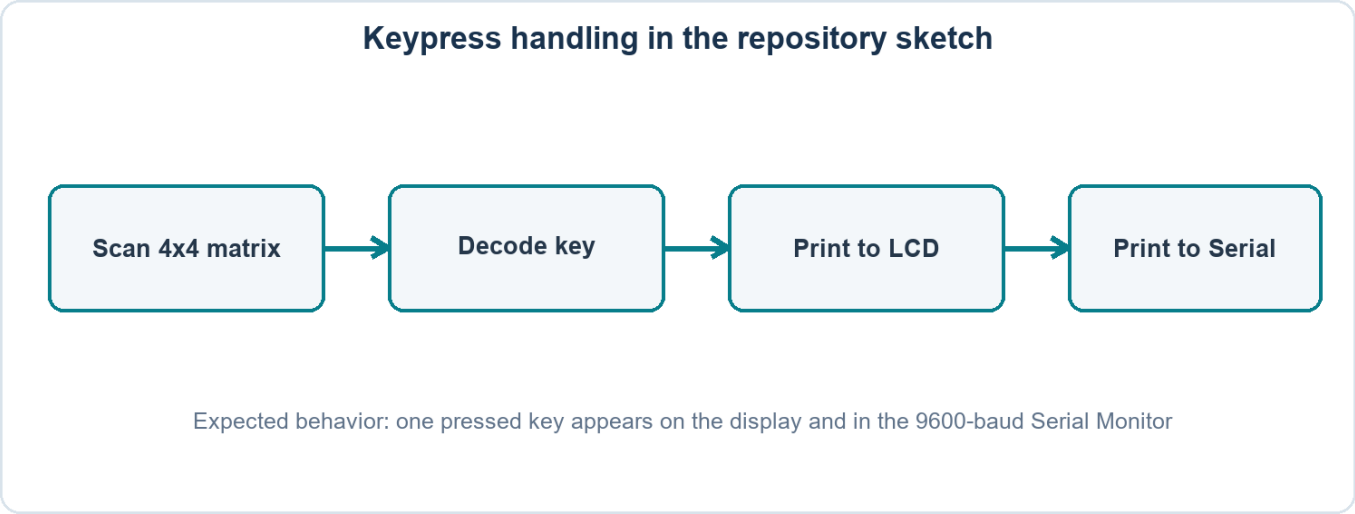


Figure 2. Intended software path from keypress to LCD and serial output.

Result and interpretation

The expected behavior is that each press appears as one character on the LCD and in Serial Monitor. No physical output or compile log for this fresh sketch is included, so this report does not claim a verified new hardware demonstration.

Limits and next validation step

Install compatible Keypad and LiquidCrystal_I2C libraries, confirm address (often 0x27), power, ground and LCD contrast, then test all keys. Wrong characters usually point to row/column order; a blank display may indicate address or power trouble.

Reproduce it

Open projects/me3202_labs/arduino_keypad_lcd/keypad_lcd.ino in Arduino IDE.

Open project code on GitHub